

Chapter 1

Theory and simulations of lattice thermal conduction

Matukumilli V. D. Prasad* and Umesh V. Waghmare†

*Theoretical Sciences Unit,
School of Advanced Materials (SAMat),
Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR),
Bangalore 560064, India.*

The linearized Boltzmann-Peierls phonon transport equation (BTE) is recently gaining popularity as a framework for describing phonon transport in solids. Here we present the developments in the simulation of lattice thermal conduction within the linearized BTE, along with the assumptions involved. The second and third-order interatomic force constants required in this analysis can be obtained from *ab initio* lattice dynamics calculations. We emphasize the shortcomings of single mode relaxation time approximation (RTA) over the full solution of BTE, which can be realized in a self-consistent fashion through an iterative approach. To the rest of this chapter, we illustrate key benefits of this approach through application to lattice dynamics and phonon transport in several materials.

1. Introduction

Heat flow is ubiquitous in nature and occurs at several scales from the thermal evolution of the terrestrial planets to the frictional dissipation in biological systems. Its implications are far-reaching. In condensed matter systems, controlling the heat flow, especially through innovative conduction mechanisms, has been crucial to advances in various technologies such as thermoelectrics, thermal management and information processing.¹ Reducing thermal conduction, for example, can enhance the thermoelectric efficiency. Understanding of the physics of heat transport in nanomaterials can help in engineering the phonon flow to realise exciting applications like

*matukumilli@jncasr.ac.in

†waghmare@jncasr.ac.in

hot-spot cooling and waste heat harvesting.^{2–5}

Calculations of phonon or lattice thermal conductivity, κ , have been carried out through either classical kinetic theory or atomistic molecular dynamics (MD). For simple systems, MD has proven to predict thermal transport accurately,⁶ it is challenging to treat complex systems due to the difficulty of obtaining accurate interatomic potentials. However, it is interesting to note that MD captures effects of anharmonicity up to all orders. A more popular approach is within the kinetic theory using linearized Boltzmann-Peierls phonon transport equation (BTE) which describes the propagation of interacting phonons in a solid.⁷ For many decades, its exact solution has remained as an intractable problem, and several methods have been developed to solve it with in some approximations.⁸ Omini and Sparavigna's iterative approach,⁹ partly due to advances in computational resources, succeeded in obtaining numerically exact solution of the full BTE.

Calculations of κ using BTE in conjunction with *ab initio* lattice dynamics have become feasible recently. As they uncover microscopic mechanisms based on phonon scattering rates and scattering phase space (which are difficult to access through experiments), they play an important role in helping establishing the chemical and physical origins of the observed behaviour of κ . The potential of these *ab initio* schemes based on third order IFCs have been established through studies on several classes of materials: Si, Ge,¹⁰ Diamond,^{11–13} GaAs,¹⁴ compound semiconductors,¹⁵ half-Heusler compounds,¹⁶ MgO,^{17,18} MgSiO₃,¹⁹ lead selenide (PbSe), lead telluride (PbTe), and their alloys,²⁰ magnesium silicide (Mg₂Si), magnesium stannide (Mg₂Sn), and their alloys,²¹ La₃Cu₃X₄ (X = P, As, Sb, Bi) compounds,²² monolayer InSe, GaSe, GaS, and alloys,²³ and boron based cubic bulk²⁴ and hexagonal 2D²⁵ compounds.

This chapter is organized as follows: We broadly discuss the theory of lattice thermal conduction in Section 2. In Section 3, we detail the simulation procedure. In Section 4, we present analysis of thermal conductivity in terms of basic vibrational properties for Al, GaAs and Diamond. The potential of BTE in understanding the impact of rattlers on κ of thermoelectric materials is highlighted Section 5. In Section 6, we cover the κ calculations of the materials that are important to understand the Earth's lower mantle. We discuss the current limitations of the BTE approach in addressing the thermal transport in complex molecular crystals in Section 7. Finally, we conclude with discussion of the main challenges in extending the phonon BTE approach to the phonon gas limit.

2. Theory

2.1. BTE under RTA

Under a small temperature gradient, ∇T , across the sample, the induced heat flux would be $\mathbf{Q} = \sum_{\lambda} \hbar \omega_{\lambda} \mathbf{v}_{\lambda} n_{\lambda}$,⁸ where n_{λ} is the phonon distribution function of phonon frequency ω_{λ} , and $\mathbf{v}_{\lambda}$ is the group velocity in x direction of phonon mode, $\lambda = \{\mathbf{q}, p\}$, where $\mathbf{q}$ is the phonon wave vector and p is the branch or polarization index. Under no other external influence, the steady state phonon distribution function can be tracked by the BTE²⁶ :

$$-\mathbf{v}_{\lambda} \cdot \nabla T \left(\frac{\partial n_{\lambda}}{\partial T} \right) + \left(\frac{\partial n_{\lambda}}{\partial t} \right)_{scatt} = 0, \quad (1)$$

which represents a balance between the diffusion due to ∇T and collective effect of all scattering processes. The primary issue in solving Eq. 1 arises from the scattering term. A popular approximation in solving the BTE is the relaxation time approximation (RTA) where the distribution function n is written as the sum of the equilibrium distribution function, n^{eq} (Bose-Einstein) and fluctuations, n^{fluc} , about equilibrium. Since $n^{fluc} \ll n^{eq}$ and assuming n^{fluc} is independent of T , we can write: $\frac{\partial n}{\partial T} \approx \frac{\partial n^{eq}}{\partial T}$, and by substituting $-\frac{n^{fluc}}{\tau}$ for the scattering term in BTE (where τ is the relaxation time) we get:

$$n_{\lambda}^{fluc} = -\tau_{\lambda} \mathbf{v}_{\lambda} \cdot \nabla T \frac{\partial n_{\lambda}^{eq}}{\partial T} \quad (2)$$

Determination of κ using the BTE within RTA is possible once the relaxation times τ 's are known. Although τ needs to be computed by other means to estimate κ , RTA facilitates writing of the complex scattering term in Eq. 1 in terms of n^{fluc} and τ .

2.2. Expression of the relaxation time

In general, phonons are scattered by boundaries, defects, isotopes, electrons and other phonons, and each of these processes involves a different scattering rate. Assuming these scattering processes are independent of each other, Matthiessen rule gives the effective scattering rate as:

$$(\tau_{eff})^{-1} = \sum_i (\tau_i)^{-1} \quad (3)$$

In case of phonon-phonon scattering, contribution of four- and higher-order phonon interactions to thermal resistance are often insignificant²⁷ and the

interactions involving three phonons dominate the scattering rates.

Using anharmonic IFCs, three-phonon scattering rates, τ_λ^{-1} , are computed using Fermi's golden rule, from the sum of all possible transition probabilities (Γ) for mode λ with modes λ' and λ'' (for absorption processes $\Gamma^+ : \lambda + \lambda' \rightarrow \lambda''$, and emission processes $\Gamma^- : \lambda \rightarrow \lambda' + \lambda''$):

$$\Gamma_{\lambda\lambda'\lambda''}^\pm = \frac{\hbar\pi}{4N} \frac{(n_\lambda^{eq} + 1) \left(n_{\lambda'}^{eq} + \frac{1}{2} \pm \frac{1}{2}\right) n_{\lambda''}^{eq}}{\omega_\lambda \omega_{\lambda'} \omega_{\lambda''}} \times \delta(\omega_\lambda \pm \omega_{\lambda'} - \omega_{\lambda''}) \times \left|\Phi_{\lambda\lambda'\lambda''}^\pm\right|^2 \quad (4)$$

where N is the number of unit cells and the energy conservation is maintained by the δ function. The three-phonon matrix element, or the strength of interaction among the three phonons, in Eq. 4, $\Phi_{\lambda\lambda'\lambda''}$, is expressed as

$$\Phi_{\lambda\lambda'\lambda''}^\pm = \sum_k \sum_{l'k'} \sum_{l''k''} \sum_{\alpha\beta\gamma} \Phi_{\alpha\beta\gamma} \left(0k, l'k', l''k''\right) \times \frac{e_{\alpha k}^\lambda e_{\beta k'}^{\pm\lambda'} e_{\gamma k''}^{-\lambda''}}{\sqrt{M_k M_{k'} M_{k''}}} e^{iq' R_{l'}} e^{iq'' R_{l''}} \quad (5)$$

where $\Phi_{\alpha\beta\gamma}$ is real-space anharmonic IFCs and takes the form as:

$$\Phi_{\alpha\beta\gamma} \left(0k, l'k', l''k''\right) = \frac{\partial^3 E}{\partial u_\alpha(0k) \partial u_\beta(l'k') \partial u_\gamma(l''k'')} \quad (6)$$

and M_k is atomic mass of k^{th} atom in Θ^{th} unit cell, $e_{\alpha k}^\lambda$ is the α^{th} component of eigen vector of phonon λ .

While analysing the anharmonic effects in thermal transport, scattering phase space has been identified recently as an important parameter along with scattering rates and anharmonic IFCs,^{22,28,29} which is the sum of the frequency-containing factors in the expression of three-phonon scattering rates (Eq. 4) and is written as:

$$W_\lambda^\pm = \sum_{\lambda', \lambda''} \frac{(n_\lambda^{eq} + 1) \left(n_{\lambda'}^{eq} + \frac{1}{2} \pm \frac{1}{2}\right) n_{\lambda''}^{eq}}{\omega_\lambda \omega_{\lambda'} \omega_{\lambda''}} \times \delta(\omega_\lambda \pm \omega_{\lambda'} - \omega_{\lambda''}) \quad (7)$$

Scattering phase space essentially measures the availability of actual energy and momentum conserving three-phonon scattering processes, assuming the strength of interaction $\Phi_{\lambda\lambda'\lambda''}$ is constant.³⁰

The scattering rates due to three-phonon processes are in the form:

$$(\tau_\lambda^{\text{RTA}})^{-1} = \frac{1}{N} \left(\sum_{\lambda', \lambda''}^+ \Gamma_{\lambda\lambda'\lambda''}^+ + \frac{1}{2} \sum_{\lambda', \lambda''}^- \Gamma_{\lambda\lambda'\lambda''}^- \right) \quad (8)$$

In addition to the three-phonon scattering process, one can also include the contribution of isotopic disorder to scattering rate, as is given by Tamura:³¹

$$\Gamma_{\lambda\lambda'} = \frac{\pi\omega^2}{2} \sum_{i \in u.c.} g(i) |\mathbf{e}_{\lambda}^*(i) \cdot \mathbf{e}_{\lambda'}(i)|^2 \delta(\omega_{\lambda} - \omega_{\lambda'}) \quad (9)$$

where g is the mass variance parameter: $g_i = \sum_j c_i^j \{(m_i^j - \bar{m}_i)/\bar{m}_i\}^2$, where m_i^j is the mass of isotope j of atom i , c_i^j is its concentration, and $\bar{m}_i$ is the average isotope mass.

2.3. From RTA and semi-empirical scheme to full solution: self-consistent calculation of the relaxation times

The three-phonon scattering processes are constrained to obey the energy and quasi-momentum conservation up to a reciprocal lattice vector:

$$\hbar\omega_{\lambda} \pm \hbar\omega_{\lambda'} = \hbar\omega_{\lambda''}, \quad \hbar\mathbf{q} \pm \hbar\mathbf{q}' = \hbar\mathbf{q}'' + a\hbar\mathbf{G} \quad (10)$$

where $\mathbf{G}$ is a reciprocal lattice vector and a is an integer whose value is zero for Normal (N) and nonzero for Umklapp (U) scattering processes. The $\pm$ signs indicate the two possible three-phonon processes: absorption and emission. At first sight, it appears that N-processes cannot be thermally resistive as the resultant phonon stays within the first BZ. However, treating them as non-resistive processes led to failure in describing experimental κ trends successfully.^{8,11,18,26} This led to development of semi-empirical models as extension of the RTA by Callaway and Holland. By highlighting the N-process as an indirect contributor to redistributing the energy among different phonons, Callaway model writes the scattering term in BTE (eq. 1) as:

$$\left(\frac{\partial n}{\partial t} \right)_{scatt} = \frac{n^N - n}{\tau_N} + \frac{-n^{fluc}}{\tau_{eff}}, \quad (11)$$

where n^N is the equilibrium distribution of phonons involved in N-processes and τ_N is the relaxation time of the N-processes. This model predicts the trends in experimental κ correctly with a few fitting parameters. Holland developed this model further by treating the longitudinal and transverse phonons separately to predict even the trends in high temperature experimental κ correctly. It is important to note that the fitting parameters and the approximations used in these semi-empirical models often obscure the fine details of phonon scattering mechanisms. Towards a parameter-free,

and a self-consistent calculation of phonon relaxation times, Omini and Sparavigna's iterative approach⁹ has been commonly used in recent years. As this iterative approach considers changes in phonon distributions due to phonon-phonon scattering events, a more rigorous treatment of anharmonic scattering became possible. Through several recent studies using this iterative approach, it is now evident that the RTA underestimates κ : by $\sim 50\%$ in diamond,¹¹ by $\sim 50\%$ in MgO,¹⁸ by $\sim 39\%$ in AlSb.¹⁵

The mode-wise relation of fluctuations in distribution to τ written explicitly in (Eq. 2) and the approximations in semi-empirical models do not allow estimation of precise τ . Now, aiming at the deviation in the distribution function due to phonon-phonon scattering itself, we can rewrite Eq. 2 as:

$$n_{\lambda}^{fluc} = -\mathbf{F}_{\lambda} \cdot \nabla T \frac{\partial n_{\lambda}^{eq}}{\partial T}, \quad (12)$$

where $\mathbf{F}_{\lambda}$ is the generalized mean free path. Unlike $\tau_{\lambda}^{\text{RTA}} \mathbf{v}_{\lambda}$, it considers the deviation in distribution functions. The resulting linearized BTE is

$$\mathbf{F}_{\lambda} = \tau_{\lambda}^{\text{RTA}} (\mathbf{v}_{\lambda} + \Delta_{\lambda}), \quad (13)$$

where Δ_{λ} (in the units of velocity) indicates the deviation (from RTA solution) in the population of a phonon mode, and $\mathbf{F}_{\lambda}$ can be regarded as a vectorial *mean free displacement*, whose direction, unlike $\tau_{\lambda}^{\text{RTA}} \mathbf{v}_{\lambda}$, may deviate from the direction of group velocity of a phonon mode.

The converged Δ_{λ} can be obtained iteratively:

$$\begin{aligned} \Delta_{\lambda} = & \frac{1}{N} \sum_{\lambda', \lambda''}^{+} \Gamma_{\lambda\lambda'\lambda''}^{+} (\xi_{\lambda\lambda''} F_{\lambda''} - \xi_{\lambda\lambda'} F_{\lambda'}) \\ & + \frac{1}{2N} \sum_{\lambda', \lambda''}^{-} \Gamma_{\lambda\lambda'\lambda''}^{-} (\xi_{\lambda\lambda''} F_{\lambda''} + \xi_{\lambda\lambda'} F_{\lambda'}) \end{aligned} \quad (14)$$

$\xi_{\lambda\lambda'}$ is a shorthand for $\omega_{\lambda'}/\omega_{\lambda}$. For both the absorption and emission three-phonon processes, these sums are over the phase space of λ' and λ'' ensuring energy and quasi-momentum conservation subject to Eq. 10. We can initiate the iterative procedure by setting $F_{\lambda'} = F_{\lambda''} = 0$, i.e. by making the zeroth-order solution equivalent to RTA ($\Delta_{\lambda} = 0$). The i^{th} iterative solution takes the $(i-1)^{\text{th}}$ iterative solution $\mathbf{F}_{\lambda}^{i-1}, \Gamma_{\lambda\lambda'\lambda''}^{\pm}$ as input and continues until it converges to arrive at the full solution of BTE in the form of $\mathbf{F}_{\lambda}$ and n_{λ} of all phonon modes. In contrast, RTA drastically

simplifies BTE by ignoring the effects of phonon-phonon scattering on the distribution functions.

Once the converged n_λ and τ_λ are obtained, the phonon thermal conductivity, κ , at temperature, T , can be calculated as the sum of the contributions of all the phonon modes, λ

$$\kappa_{\alpha\beta} = \frac{1}{N\Omega} \sum_{\lambda} \frac{\partial n_{\lambda}}{\partial T} (\hbar\omega_{\lambda}) v_{\lambda}^{\alpha} v_{\lambda}^{\beta} \tau_{\lambda}, \quad (15)$$

or implemented in the form:

$$\kappa_{\alpha\beta} = \frac{1}{k_B T^2 N\Omega} \sum_{\lambda} n_{\lambda}^{eq} (n_{\lambda}^{eq} + 1) (\hbar\omega_{\lambda})^2 v_{\lambda}^{\alpha} F_{\lambda}^{\beta} \quad (16)$$

where N , Ω are the number of q points sampled in integration over Brillouin zone and unit cell volume respectively.

We note here that the definition of τ will be compromised beyond the RTA as the deviation from n^{eq} became the fundamental parameter in the full BTE solution. Thus, the nature of final quantities τ , and MFP, Λ are no longer the same as they are described within the kinetic theory. However these are embedded in the $\mathbf{F}_{\lambda}$'s to calculate κ . When the deviation of these τ , and Λ from those within the kinetic theory is large, the RTA fails badly.

3. Simulation Procedure

We present here a procedure in which harmonic and anharmonic IFCs are obtained using first-principles calculations within density functional theory (DFT) and density functional perturbation theory (DFPT) (e.g. as implemented in Quantum Espresso³² code). We illustrate it through applications to Al, GaAs and Diamond. In DFT calculations, we employed a generalized gradient approximation (GGA) to the exchange correlation energy functional and used norm-conserving pseudo-potentials to treat the interactions between ionic cores and valence electrons, and a plane wave basis truncated with an appropriate energy cutoff in the representation of the Kohn-Sham wave functions. The discontinuity in the occupations numbers of electronic states near the gap or Fermi level were smeared with Fermi-Dirac distribution function with a suitable broadening parameter.

To use Fourier transform for obtaining IFCs at any arbitrary q -point, harmonic IFCs are determined at q -vectors on a mesh using DFPT linear response. In calculation of anharmonic third order IFCs, we employed finite difference approach and used periodic supercells. ShengBTE's *third-order.py* code is used to generate displaced configurations of $4 \times 4 \times 4$

supercell with upto 5th nearest neighbour interactions and the resulting atomic forces are computed in QE. We need these first-principles harmonic and anharmonic IFCs in the ShengBTE³³ code to solve BTE using iterative method.⁹

4. Illustration of computation of Lattice thermal conductivity

4.1. Aluminum

Extensive studies have been devoted to calculate the κ of various nonmetals which lead to the better understanding of phonon thermal transport in semiconductors and insulators.^{10,11} However, relatively less attention has been received by metals due to dominant electronic contribution to thermal transport in many of the metals. Though, the coupled el-ph heat transfer can be addressed through two-temperature model in metals, which is a phenomenological in nature, it fails in the case of semiconductors due to their heterogenous nature of el-ph interactions. Very recently, phonon-limited thermalization has been identified in semiconductors using a generalized two-temperature model³⁴ wherein both electrons and phonons do not remain in distinct thermal equilibria. Such simultaneous treatment of electronic and phononic subsystems could lead to the computation of evolution of nonequilibrium phonon distributions emerged due to electronic excitations and their effects on the further thermal relaxation of electrons.

Aluminum is an interesting example for the study of phonon thermal transport. Since Al is a free-electron-like metal, which can be demonstrated from the quadratic behaviour of its electronic DOS, it is likely that el-ph coupling can significantly affect the phonon thermal transport, κ . Unlike Ni and Pt, the electron-phonon scattering rates, τ_{el-ph}^{-1} in Al are insignificant in comparison to its phonon-phonon scattering rates, τ_{ph-ph}^{-1} . Wang et al.,³⁵ calculated the thermal conductivity and phonon scattering rates of several metals, including Al. Among the studied metals, they found that τ_{el-ph}^{-1} is negligible in Cu, Ag, Au and Al at room temperature whereas it is found to be significant in Ni and Pt. Jain and McGaughey³⁶ also studied the thermal transport in Al, Ag, and Au by considering both electron-phonon and phonon-phonon scattering.

Phonon dispersion of Al in the high symmetry directions in the BZ calculated in this work is shown in Figure 1. The calculated κ (Figure 2 (a)) exhibits reduction with temperature due to enhanced ph-ph anharmonic

scattering as evident in Figure 2 (b). The BTE is solved on three different grids: $15 \times 15 \times 15$, $20 \times 20 \times 20$ and $30 \times 30 \times 30$ and values of κ differ no more than 3% at low temperatures, which indicates that the $15 \times 15 \times 15$ grid is sufficient to give converged κ values. The calculated estimates of κ are in reasonable agreement with the theoretical estimates of Wang et al.³⁵

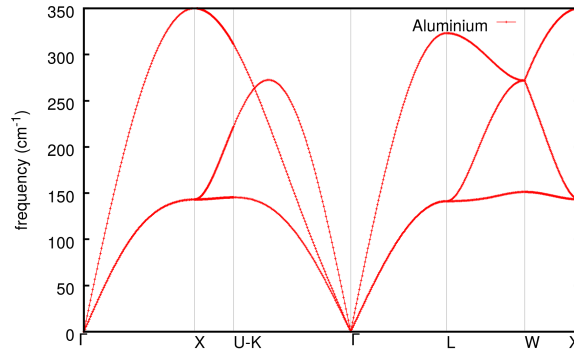


Fig. 1. Phonon dispersion of Al.

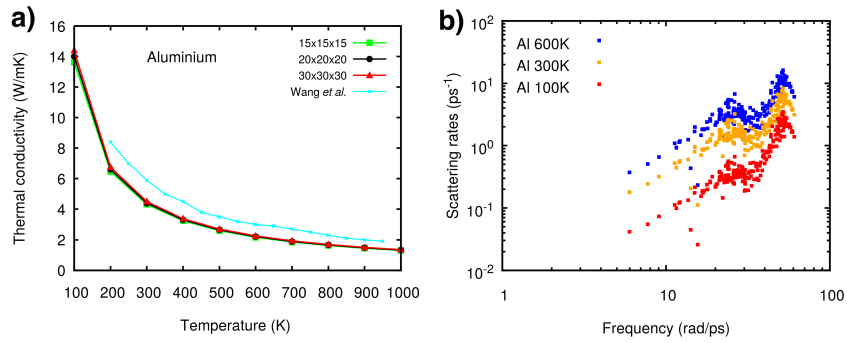


Fig. 2. Calculated intrinsic lattice thermal conductivity of Al as a function of temperature from the full converged solution at three different grids: $15 \times 15 \times 15$ (green squares), $20 \times 20 \times 20$ (black circles) and $30 \times 30 \times 30$ (red triangles), compared with the κ results of Wang et al.³⁵ (cyan crosses) involving ph-ph interactions (a). (b) Anharmonic ph-ph scattering causes the $1/T$ behavior of κ -T relation. Anharmonic phonon scattering rates in Al at three different temperatures: 100 K (red), 300 K (orange), and 600 K (blue) showing stronger scattering at elevated temperatures.

4.2. Diamond

κ of Diamond can be higher than 3000 W/m-K, which is the highest among the known materials.^{37–39} Such an extremely high κ is possible due to: (i) high bond strength and low atomic mass giving large acoustic velocities (ii) weak ph-ph umklapp processes. The weak U processes in Diamond made it difficult to calculate κ using BTE within RTA. At room temperature, κ estimated with iterative solution is 50% higher value than κ estimated within RTA.¹¹ In semiconductors, this difference is only $\sim 10\%$ due to the presence of strong U scattering. Such particularly weak U scattering in Diamond in comparison to semiconductors like Si and Ge has also been indicated from the results of available three-phonon scattering phase-space.³⁰ For Diamond, the three-phonon scattering phase-space is surprisingly small in comparison to these semiconductors. Several other examples where RTA underestimates κ includes AlSb by 39%,⁴⁰ MgO by 30%.¹⁸

In Figure 3 (a) we present the calculated κ for Diamond by iteratively solving the BTE. These values are in good agreement with the earlier experimental^{38,39} and theoretical²⁴ results. RTA's under-prediction of κ w.r.t the full solution is also another signature of the correctness of our calculations. From these trends, it is clear that κ from the iterative solution showed 43% higher value than κ obtained within RTA at 200 K. At 500 K, it is only 32% higher.

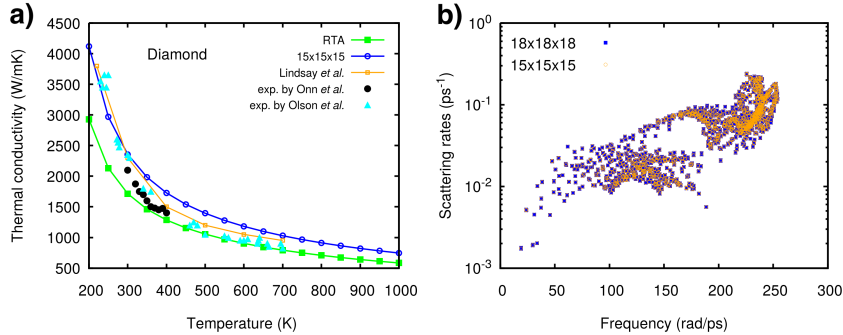


Fig. 3. Intrinsic lattice thermal conductivity of Diamond as a function of temperature within RTA (green squares) and with full converged solution (blue circles) at $15 \times 15 \times 15$ grid. Our results are compared with those of experiments by Onn et al.³⁸ (black circles) and Olson et al.³⁹ (cyan triangles), and *ab initio* study by Lindsay et al.²⁴ (orange squares) (a). (b) Anharmonic phonon scattering rates in Diamond computed as a function of frequency obtained on two different q-point grids: $15 \times 15 \times 15$ (orange circles), $18 \times 18 \times 18$ (blue squares).

In comparison with κ of Al (from Figure 2(a)), Diamond exhibits extremely large κ throughout the studied temperature range. Besides, in the both the cases, the decreasing trend of κ with temperature is due the enhanced anharmonic ph-ph scattering. Calculated anharmonic phonon scattering rates of Diamond (see Figure 3 (b)) are two orders of magnitude lower than the scattering rates of Al, which explains extremely high κ of diamond.

Several other scenarios have also uncovered through *ab initio* simulations of thermal transport where κ can even be higher than Diamond.²⁴ Studies of Lindsay et al., on κ of a class of boron based cubic compounds (BN, BP, BAs, and BSb) above room temperature, showed that BAs surprisingly showed higher κ than Diamond. This unusually high κ has been explained, from the phonon dispersion of BAs, as originating from the interplay of: (i) a large frequency gap between acoustic and optic phonons and (ii) the bunching of the acoustic branches. This highlights the capabilities of advanced computational techniques for determination of thermal transport properties, and have opened up novel $\omega(q)$ features beyond the conventional criteria for designing high κ materials.

4.3. GaAs

With miniaturisation of electronic devices, understanding of thermal properties of semiconductors has become crucial. Thermal transport in group IV, III-V and II-VI bulk semiconductors has been investigated using *ab initio* approaches based on the solutions of BTE.^{30,40} GaAs has been extensively studied and widely used in opto- and micro-electronic devices. Hot phonon effects, wherein optical phonons exists in highly nonequilibrium populations in response to the absorbed photo-excited electrons,^{41,42} has been attributed to the reduction of carrier cooling rates. Knowledge of phonon relaxation times could help in better understanding of hot phonon effects on the optoelectronic device performance.¹⁴ Semiconductor superlattice structures, where κ can be tuned efficiently, are promising materials especially as thermoelectrics.⁴³ In Luckyanova et al.,’s work⁴³ based on *ab initio* thermal transport calculations using BTE, the frequency-mean free path relation has been used to explain the coherent nature of phonon transport in GaAs/AlAs superlattice structures.

Figure 4 (a), shows the calculated κ values for GaAs. In Figure 4 (b), we present a comparison of anharmonic scattering rates of GaAs and Al at different temperatures. Overall scattering rates of GaAs are an order of

magnitude smaller than those of Aluminum.

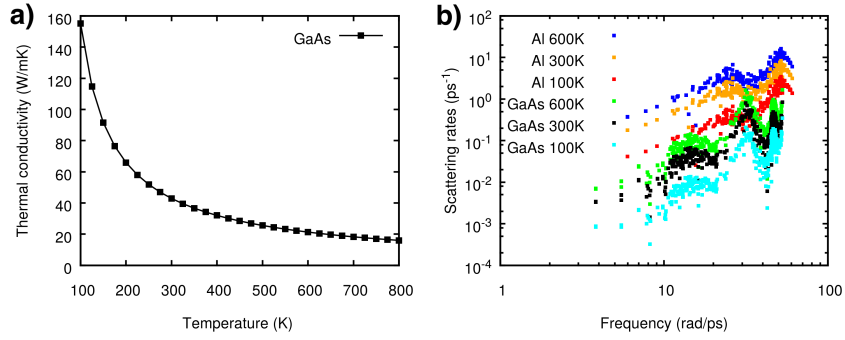


Fig. 4. The calculated intrinsic lattice thermal conductivity of GaAs as a function of temperature (a). (b) Anharmonic phonon scattering rates in GaAs as a function of frequency in comparison with Al at three different temperatures.

The normalized cumulative contribution to κ , which indicates a fraction of κ by the phonons of MFP below a certain value, is shown in Figure 5. Comparison between cumulative κ of Al and GaAs provides a clear picture on the thermal transport by phonons with different MFPs. In Al, 90% of κ is contributed by phonons with MFPs less than 10 nm, 90% of κ of GaAs is contributed by its phonons with MFPs less than 1 μm .

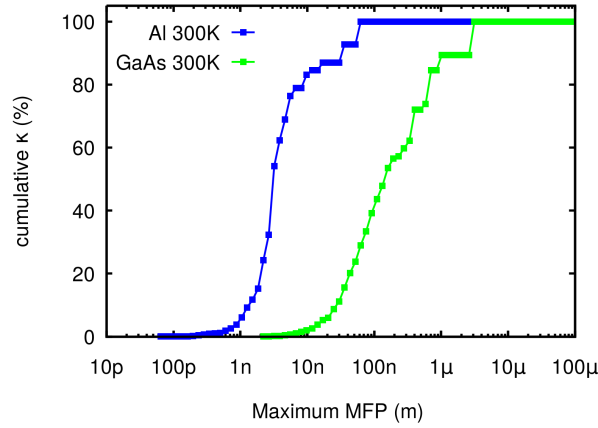


Fig. 5. Cumulative thermal conductivity of Al (blue) and GaAs (green) at 300 K. κ values are normalized by their final values ($\kappa_{Al} = 4.12$ W/m K and $\kappa_{GaAs} = 42.89$ W/m K at 300 K) to bring them on to the same scale for comparison.

5. Impact of rattlers on κ of thermoelectric clathrates

A key aspect of designing a high-performance thermoelectric material is to minimize its κ to the theoretical minimum. In general, materials with cage-like crystal structures (clathrates and skutterudites) exhibit thermal conductivity close to their theoretical minimum and, initially, this has been attributed to Einstein rattling mode.^{44,45} In other classes of materials like in Zintl type TlInTe_2 ,⁴⁶ along with clathrates and skutterudites,^{44,45} the suppression of κ by rattling modes has been found as an exciting concept, which actually hinted the possibility of nontraditional phonon scattering scenarios. Intense research efforts have been devoted to identify efficient routes (such as by intrinsic means) to attain better performing thermoelectric materials by exploiting the material chemistry and bonding nature.⁴⁷ Though probing the rattler modes experimentally,^{48,49} by means of inelastic X-ray and inelastic neutron spectroscopies, could provide preliminary understanding of phonon transport in these guest-host structures, theoretical approaches like *ab initio* lattice dynamics based BTE, with a detailed information on phonon scattering dynamics,²⁷ can further provide insights into chemical and physical origins of ultra-low κ .

Using a simple Born-von Karman (BvK) model, which describes the lattice as a chain of atoms connected by harmonic springs, Toberer et al.,⁵⁰ analysed the effect of mass ratio, number of atoms on κ . With a modified version of this BvK model where the stiffness of springs altered such that the behaviour of loosely bound guest atom can be incorporated, they were able to investigate the guest-host like compounds. Results of analysis of the resonant scattering model are presented in Figure 6, which emphasise that the information on frequency-dependent group velocities and scattering rates are vital to unravel the coexisting effects on κ .

5.1. $\text{Ba}_8\text{Ga}_{16}\text{Ge}_{30}$ (Ref. 27)

Here, we highlight the role of anharmonic phonon interaction based calculation of conductivity in clarifying the impact of a rattler, by discussing results of the study by Tadano et al.²⁷ on $\text{Ba}_8\text{Ga}_{16}\text{Ge}_{30}$. By separately calculating phonon properties of Ba filled (BGG) and empty $\text{Ga}_{16}\text{Ge}_{30}$ (GG) clathrate structures (see Figure 7), they observed a tenfold reduction in phonon scattering rates due to rattler dynamics. Importantly, from the analysis of anharmonic scattering rates (see Figure 8), they show that the effects of a rattler on the phonon modes is not restricted to a certain

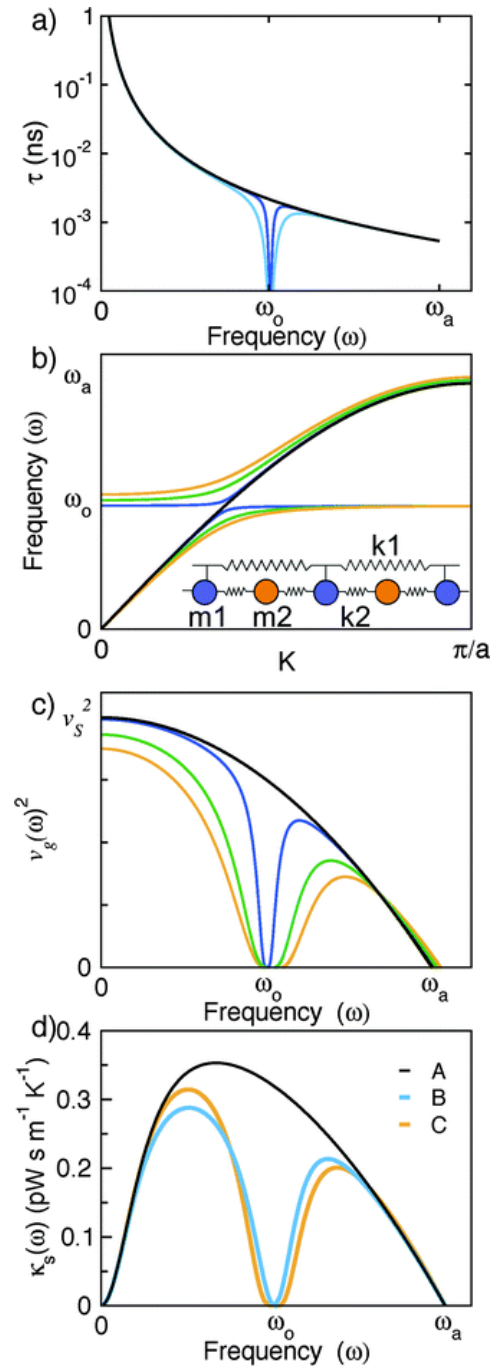


Fig. 6. The resonant scattering model targets phonons near ω_0 (a). BvK phonon dispersions for a stiff framework (m_1, k_1) and loosely bound guest atoms (m_2, k_2) (b). Increased k_2 stiffness results in increased coupling (extent of avoided crossing) between the framework and guest modes. c) The avoided crossing reduces $v_g(\omega)^2$ in the vicinity of ω_0 . d) $\kappa_s(\omega)$ for an empty BvK framework, using Umklapp and boundary scattering terms (curve A). Including resonant scattering reduces $\kappa_s(\omega)$ near ω_0 (curve B). If instead, the effect of coupling on $v_g(\omega)$ is accounted for, a similar reduction is observed (curve C). Reprinted with permission from Ref.50. Copyright 2011 The Royal Society of Chemistry.

frequency range and thus argued that the phonon scattering in BGG is nonresonant.

Conventionally, ultra-low κ in guest-host structures has been explained by *resonant* scattering mechanism,^{51,52} where only a few phonon modes are affected by the rattler, leading to disordered lattice dynamics, which is popularly termed as *phonon glass-electron crystal*. Rattler effect is understood by observing: (i) a shallow potential implying a loose bonding which cause weak restoring forces on the vibrating filler, and (ii) lack of phase coherence of the vibrational motion.⁴⁸

As implications of crystallographic heterogeneity on anharmonicity are more complex, such *rattling mode* explanation failed in explaining the low thermal conductivity in certain compounds. Other mechanisms such as (i) influence of *flat avoided-crossing bands of filler modes* in phonon scattering, (ii) interplay of anharmonicity and scattering phase space also have been identified in explaining low thermal conductivity of skutterudites. In La- and Ce-filled $\text{Fe}_4\text{Sb}_{12}$ skutterudites⁴⁸ it has been observed through neutron spectroscopy experiments and *ab initio* calculations, that the filler vibrational modes are *coherently* coupled with the host-lattice dynamics. While this experimental work initiated the counter argument on the resonant nature of the rattler, only the theoretical studies based on BTE have been able to confirm it later by providing a detailed understanding of the phonon behaviour. Tadano et al.'s work²⁷ is one such example.

Phonon dispersion and density of states (DOS) of BGG and GG calculated by Tadano et al.,²⁷ are reproduced in Figure 7(a). The rattling modes of Ba(2) atoms can be seen in 28-34 cm^{-1} range along with the avoided crossing of LA modes. From the DOS, Ba(2) induced modes (in blue line) can also be seen at 55 cm^{-1} . These rattling mode frequencies are in good agreement with experiments.⁵³ Firstly, they have calculated the participation ratio, P_q , for each phonon mode, q , as $P_q = [\sum_{\alpha} |\mathbf{u}_{\alpha}(q)|^2]^2 / N_{\alpha} \sum_{\alpha} |\mathbf{u}_{\alpha}(q)|^4$ where $\mathbf{u}_{\alpha}(q) = M_{\alpha}^{-1/2} \mathbf{e}_{\alpha}(q)$, M is atomic mass of atom α and $\mathbf{e}$ is the eigen vector. From the P_q values, they found that the localization of guest atoms is significant in the rattling modes around 28-34 cm^{-1} which is in accordance with the Einstein-rattler picture—a popular notion. However, this participation ratio is an indirect measure of the effect of Ba rattler. To have a direct measure, they used *ab-initio* lattice dynamics along with BTE to calculate κ , phonon line width and phonon scattering phase space. The harmonic and cubic IFCs were calculated *ab-initio* only for BGG, as the GG structure is fictitious. The

trick they have used to generate IFCs for GG from the IFCs of BGG is that they turned off the interactions between the host lattice and guest atoms. At 100 K, calculated κ of GG structure is 20 times greater than that of BGG. Estimated κ of BGG at 100 K is 1.35 W/m-K, which matches well with experimental values (1.5 W/m-K for p-type and 1.9 W/m-K for n-type).⁵⁴

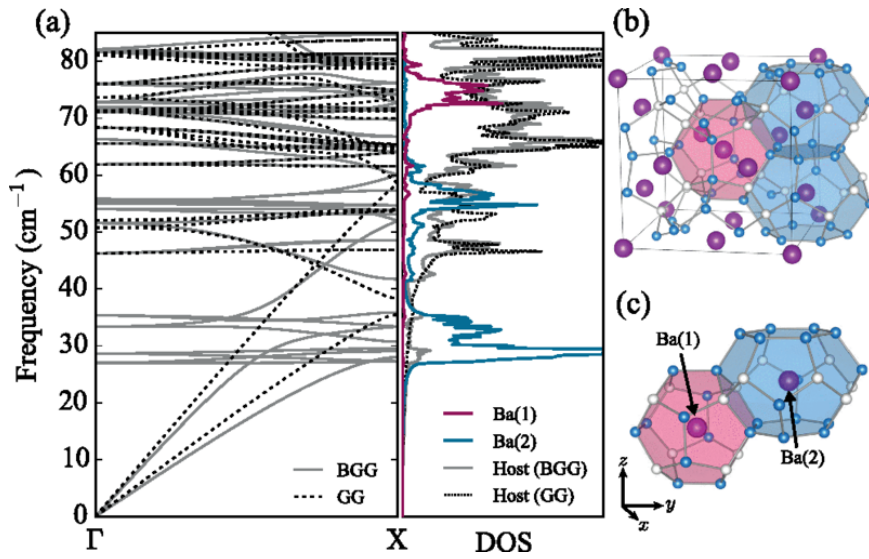


Fig. 7. (a) Calculated phonon dispersion and projected phonon DOS of BGG (solid lines) and GG (dotted lines). (b) Crystal structure of BGG. Ba, Ga, and Ge atoms are represented by magenta, white, and blue spheres, respectively. (c) Representation of two inequivalent Ba rattlers in dodecahedral (red) and tetrakaidecahedral (blue) polyhedra. Reprinted with permission from Ref.27. Copyright 2015 American Physical Society.

Analysing the scattering rates τ^{-1} of these structures, they obtained important insights: the enhancement of τ^{-1} due to rattler is not limited to any specific frequency region, it is seen in the entire frequency range, profoundly below 120 cm^{-1} . Firstly, this 10-fold increase in τ^{-1} explains the 20-fold reduction in κ due to Ba rattler. Secondly, the change of τ^{-1} in entire region is in contradiction with the resonant scattering picture where such effect is localised in frequency domain. Despite these insights from the trends in τ^{-1} , the reason for the increase in τ^{-1} in BGG is still unknown and even the results on three-phonon scattering phase space, $W^{\pm}$, (inset of Figure 8) could not explain it as the increase in $W^{\pm}$ is only by a factor of 2,

which is inadequate in explaining the 10-fold increase of τ^{-1} . Interestingly, the trends in three-phonon matrix elements (in other words, the strength of interaction among the three phonons), $\Phi_{\lambda\lambda'\lambda''}$, as in Eq. 5, appears promising. A significantly higher values of $\Phi_{\lambda\lambda'\lambda''}$ in BGG in comparison to GG has been observed in the frequency ranges precisely corresponding to the frequencies of the localised rattling modes (Figure 7). $\Phi_{\lambda\lambda'\lambda''}$ being a measure of anharmonicity in the system, this increase in $\Phi_{\lambda\lambda'\lambda''}$ is a clear indication of effects of rattler in BGG. This work by Tadano et al.,²⁷ could certainly showcase the enormous potential of the anharmonic lattice dynamics with BTE in understanding the mechanism of thermal transport.

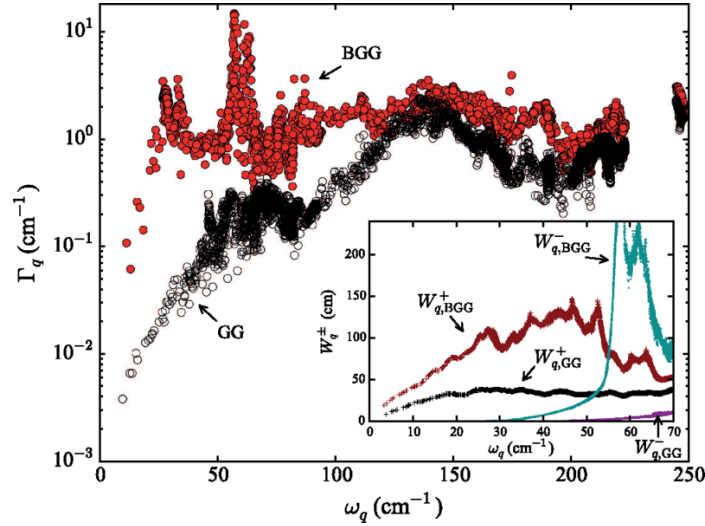


Fig. 8. Calculated phonon linewidth of BGG (filled circles) and GG (open circles) at 300 K. Inset: Energy- and momentum-conserving phase space $W_q^\pm$ in the low-energy region at 300 K. Reprinted with permission from Ref.27. Copyright 2015 American Physical Society.

5.2. CoSb₃ and BaCo₄Sb₁₂ (Ref. 29)

Li and Mingo²⁹ estimated κ for CoSb₃ and BaCo₄Sb₁₂ skutterudites using *ab initio* lattice dynamics. This is another study where BTE helped in settling the controversial explanation of rattling mechanism. From the comparison between scattering rates of Ba filled skutterudite (BaCo₄Sb₁₂) and empty skutterudite (CoSb₃), they demonstrated that filler causing an

increase in scattering rates by a factor of 1.7. Firstly, this enhanced scattering through out the frequency range ruled out the puzzling resonant nature of rattlers; in the resonant rattler picture such enhancement of scattering is limited to narrow range of frequencies. To uncover the precise role of rattler in τ^{-1} , they calculated the τ^{-1} of $\text{BaCo}_4\text{Sb}_{12}$ by *turning off* the interactions between Ba and the host, and found τ^{-1} is still significantly higher than empty skutterudite CoSb_3 (see Figure 9). That indicates that the enhanced scattering in $\text{BaCo}_4\text{Sb}_{12}$ is not caused by the presence of filler alone. They also checked the role of third-order IFCs in the enhanced scattering even in the absence of guest-host interactions. For that, they used the CoSb_3 's third-order IFCs in calculating τ^{-1} in $\text{BaCo}_4\text{Sb}_{12}$. They did not see much difference in the scattering trends compared to the previous case (see bottom part of Figure 9). Thus, neither the third-order IFCs nor the guest-host interactions cause the enhanced scattering in filled skutterudite. Another possibility is the second-order IFCs which essentially contribute to the scattering rates through frequencies. The three-phonon scattering phase space, $W^\pm$, (see Eq. 7) includes frequencies from the transition probabilities equation (see Eq. 4). After analysing the $W^\pm$ trends, they concluded that the depressed phonon spectrum is the cause for the enhanced scattering in filled skutterudites.

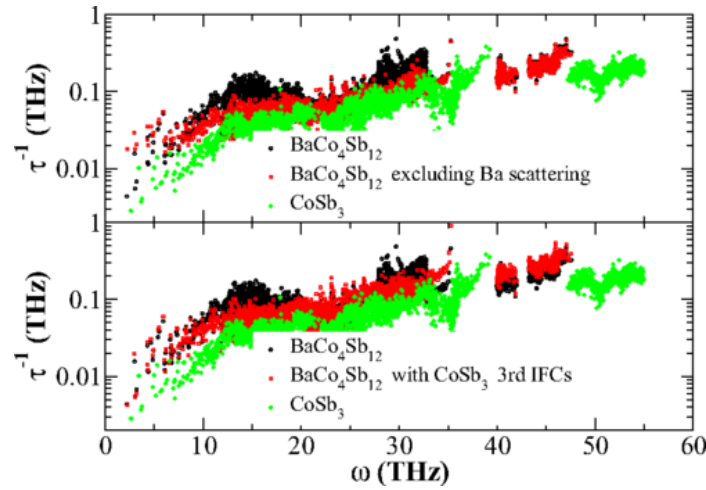


Fig. 9. Top: Anharmonic scattering rates of $\text{BaCo}_4\text{Sb}_{12}$, compared with those calculated by excluding Ba-related scattering. Bottom: Anharmonic scattering rates of $\text{BaCo}_4\text{Sb}_{12}$, compared with those calculated by using the third-order IFCs of CoSb_3 . Reprinted with permission from Ref.29. Copyright 2014 American Physical Society.

Indeed, such thorough investigation became possible with the valuable and multifaceted information provided by *ab initio* based BTE method.

6. Thermal conductivity of the Earth's lower mantle

Thermal structure and evolution of the Earth's interior primarily depends on the heat flow through the Earth's core-mantle boundary, which involves exciting science of thermal boundary layer in terms of heat conduction.⁵⁵ As the conduction plays a major role in the heat transport across the core-mantle boundary, an understanding of planetary heat transport requires knowledge of κ as functions of pressure, absolute temperature and chemical composition. This core-mantle thermal structure also has strong correlations with the geomagnetic field morphology.⁵⁶ As the primary constituents of core-mantle boundary layer, polycrystalline aggregate of silicate perovskite ((Fe,Al)-bearing MgSiO_3) and ferropericlase ((Mg,Fe)O) have attracted numerous experimental^{57–59} and theoretical investigations.^{17–19,60,61} Earth's core conditions in terms of extreme pressures and temperatures pose technical challenges in measuring κ ^{57,58,62,63} and thus the extrapolated parameters from ambient conditions have been used in the models to predict behaviour of Earth's interior.^{64,65} In the initial studies, phonon group velocities are approximated as sound velocities and phonon lifetimes at extreme conditions are approximated with lattice thermal expansion and/or Grüneisen parameters which are equilibrium thermodynamic properties.

In the first attempt to calculate κ of MgO, molecular dynamics simulations and Green-Kubo theory has been used.⁶⁷ Due to the failure of the interatomic potentials, this study underestimated the κ by an order of magnitude. Later, de Koker⁶⁸ used first-principles molecular dynamics along with phonon spectral density approach to arrive at phonon lifetimes, a crucial parameter in estimation of κ . First study that used anharmonic lattice dynamics to calculate κ is by Tang and Dong,¹⁷ which marked a significant improvement in modeling κ because of the accuracy in κ it predicted of MgO. Figure 10 shows the calculated κ values of MgO as a function of pressure and temperature. They considered a hot and cold geotherm based on experimental constraints⁶⁶ to give a direct estimation of κ of MgO at the lower-mantle conditions. These hot and cold geotherms correspond to the whole-mantle convection and partially-layered convection respectively, and provide a reasonable temperature bound for the lower mantle. At the same depth, there is a $\sim 5\text{--}10$ W/mK difference in κ across the hot and cold

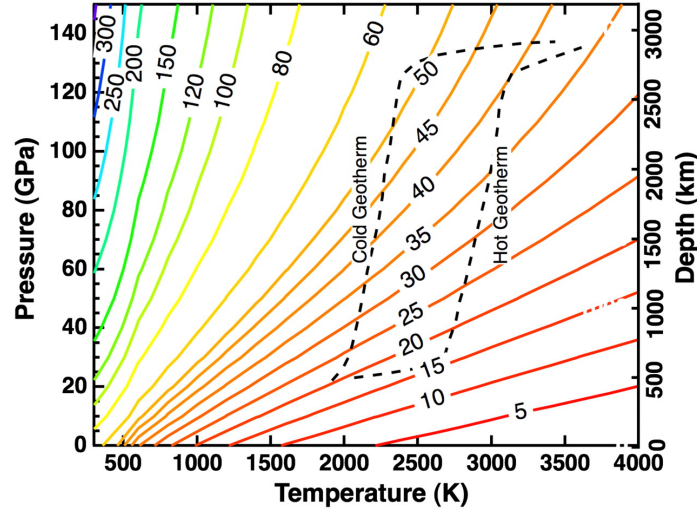


Fig. 10. Pressure and temperature contour plot of the lattice thermal conductivity of MgO. Dashed Lines are estimated cold and hot geotherms from experiments (Ref. 66). Reprinted with permission from Ref.17. Copyright 2010 National Academy of Sciences.

geotherms. There is a strong dependence of κ on depth. Though this work based on first-principles calculations succeeded in improving the accuracy of κ estimates of metal oxide materials, it is still far from the realistic values of κ .

The major concern is the iron enrichment in these mantle minerals which drastically affect the physical and chemical properties of the core-mantle boundary.^{69,70} Recently, there have been studies to understand the effects of iron on κ of Earth's deep mantle and its implications for mantle dynamics.^{57,71} They found even a small percentage of Fe can strongly reduce κ . Similar trends have been found even for larger concentrations of Fe in ferropericlase. Substitution of Al, instead of Fe, in perovskite also led to a significant reduction of κ .⁵⁷ There have been no theoretical efforts, using lattice dynamics and BTE, to calculate κ of mantle minerals enriched with small % of Fe or Al.

The other concern, along with the compositional disorder, is the extreme conditions in terms of high pressures and high temperatures, where experimental data is highly scarce and accuracy of theoretical predictions is very poor. For MgO, Dekura and Tsuchiya¹⁸ calculated κ at extreme conditions (up to 150 GPa and 4000 K) using an iterative solution of BTE and compared with RTA results (see Figure 11). They find that RTA underes-

estimates κ of MgO, by about 30%, at both ambient and extreme conditions. Similar to the case of Diamond, U phonon scattering is weak in MgO and thus iterative solution of BTE is essential to determine κ accurately.

Iterative solution of BTE can be a potential theoretical tool in improved estimation of κ at Earth's mantle conditions, and can be extended to the other mineral compounds.

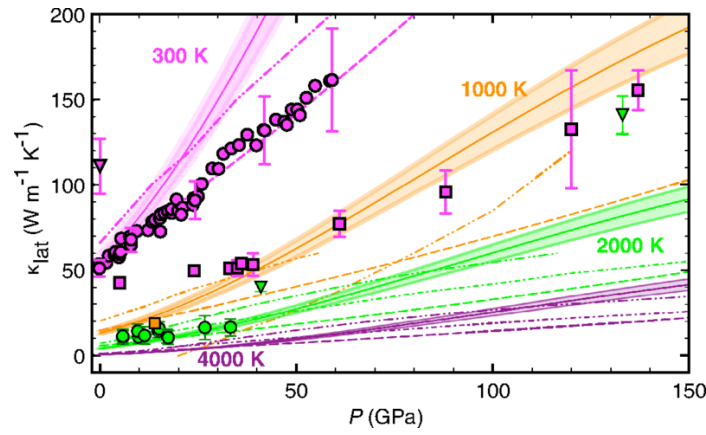


Fig. 11. Calculated κ of MgO with the fully converged solutions to the linearized BTE (solid lines) as a function of pressure P at several T from 300 to 4000 K with $1\sigma = 8\%$ confidence bands (shaded regions). The results of previous studies based on *ab initio* ALD calculations with the isotopic correction (dashed double-dotted lines) [17], equilibrium (dash-dotted lines) [72] and nonequilibrium MD (dashed lines) [73], and experiments conducted using single-crystal samples (pink and green circles for 300 and 2000 K, respectively) [62,63] and polycrystalline samples (pink and orange squares for 300 and 1000 K, respectively) [57,58] are provided. The results of classical MD simulations (pink and green triangles for 300 and 2000 K, respectively) [74] are also depicted. Reprinted with permission from Ref.18. Copyright 2017 American Physical Society.

7. Molecular crystals: Limitations of BTE approach

Phonon behavior in energetic molecular crystals such as RDX, HMX, and TATB is of tantamount importance, especially, to understand the ignition mechanism. For atomic crystals, BTE serves well by providing detailed phonon scattering mechanisms. In molecular crystals such as RDX and TATB energetic materials, it is computationally expensive to deal with large unit cells and it is a long calculation to obtain phonon dispersion. Thus the task of obtaining scattering rates and κ of molecular crystals by solving

BTE is computationally not feasible. However, there have been attempts, using other theoretical approaches, to understand the role of phonon scattering in the process of ignition of energetic molecular crystals. In order to calculate κ for energetic material, Long and Chen⁷⁵ defined *phonon stress* from individual phonon momenta and obtained phonon-phonon interactions. Based on this, they identified key vibrational modes relevant to heat exchange processes.

Even for atomic crystals, as the crystal complexity increases, inherent anharmonicity starts to dominate and it manifests as weakly dispersed phonon branches, leading to localization of heat.

8. Conclusions and Outlook

We have presented computational studies of thermal transport behaviour of several materials to highlight the key features of phonon BTE approach based on *ab initio* lattice dynamics calculations. We started with a detailed presentation of the theory and its developments, and presented our theoretical results for thermal conduction in simple crystals like Al, Diamond and GaAs. Moving to the thermoelectrics, we have reviewed how the theoretical studies using BTE helped in demystifying the nature of rattle mechanism—one of the intricate mechanisms of ultra-low κ in thermoelectrics. We have also highlighted the role of theoretical studies using BTE in accurately predicting thermal conductivity of materials under extreme conditions with examples of prominent minerals of Earth's lower-mantle. As demonstrated here, phonon BTE along with first-principles lattice dynamics is an exceptional tool, which is yet to be utilised in many other areas and materials.

Nevertheless, extending the phonon BTE beyond the phonon gas limit, would be an important step forward. The theoretical foundation of the phonon BTE is rooted in the phonon gas model,^{1,26} where vibrational modes are weakly interacting such that populations of each phonon follow the single-particle Bose-Einstein distribution at equilibrium. This picture is not true in glassy and amorphous systems. Recently, two-channel model has been proposed,⁷⁶ where an extra component from the uncorrelated oscillations of vibrational energy were found to be necessary to explain the deviation of phonon BTE estimate of κ from experiments (at high temperatures) in crystalline Ti_3VSe_4 . Beyond the phonon gas limit, *vibration Fokker-Planck equation theory* has been proposed to describe the stochastic fluctuation and relaxation processes of lattice vibrations at a wide range

of conditions.⁷⁷ Such a theory makes multiple-mode correlation functions accessible and has the potential to possibly change the paradigm of thermal physics in the near future.

9. Acknowledgements

UVW thank support from a J. C. Bose National Fellowship of the department of science and technology of Government of India, a Sheikh Saqr Fellowship, and IKST-Bangalore.

References

1. G. Chen, *Nanoscale Energy Transfer and Conversion*. Oxford University Press (2005).
2. C. Melnick and M. Kaviany, Phonovoltaic. i. harvesting hot optical phonons in a nanoscale $p-n$ junction, *Phys. Rev. B.* **93**, 094302 (Mar, 2016). doi: 10.1103/PhysRevB.93.094302. URL <https://link.aps.org/doi/10.1103/PhysRevB.93.094302>.
3. C. Melnick and M. Kaviany, Phonovoltaic. ii. tuning band gap to optical phonon in graphite, *Phys. Rev. B.* **93**, 125203 (Mar, 2016). doi: 10.1103/PhysRevB.93.125203. URL <https://link.aps.org/doi/10.1103/PhysRevB.93.125203>.
4. C. Melnick and M. Kaviany, Phonovoltaic. iii. electron-phonon coupling and figure of merit of graphene:bn, *Phys. Rev. B.* **94**, 245412 (Dec, 2016). doi: 10.1103/PhysRevB.94.245412. URL <https://link.aps.org/doi/10.1103/PhysRevB.94.245412>.
5. A. Seif, W. DeGottardi, K. Esfarjani, and M. Hafezi, Thermal management and non-reciprocal control of phonon flow via optomechanics, *Nature Communications.* **9**(1), 1207 (2018). ISSN 2041-1723. doi: 10.1038/s41467-018-03624-y. URL <https://doi.org/10.1038/s41467-018-03624-y>.
6. B.-L. Huang and M. Kaviany, Ab initio and molecular dynamics predictions for electron and phonon transport in bismuth telluride, *Phys. Rev. B.* **77**, 125209 (Mar, 2008). doi: 10.1103/PhysRevB.77.125209. URL <https://link.aps.org/doi/10.1103/PhysRevB.77.125209>.
7. R. Peierls, Zur kinetischen theorie der wrmeleitung in kristallen, *Annalen der Physik.* **395**(8), 1055–1101 (1929). ISSN 1521-3889. doi: 10.1002/andp.19293950803. URL <http://dx.doi.org/10.1002/andp.19293950803>.
8. J. M. Ziman, *Electrons and Phonons: The Theory of Transport Phenomena in Solids*. Oxford University Press, London (1960).
9. M. Omini and A. Sparavigna, An iterative approach to the phonon boltzmann equation in the theory of thermal conductivity, *Physica B: Condensed Matter.* **212**(2), 101 – 112 (1995). ISSN 0921-4526. doi: [https://doi.org/10.1016/0921-4526\(95\)00000-0](https://doi.org/10.1016/0921-4526(95)00000-0).

- org/10.1016/0921-4526(95)00016-3. URL <http://www.sciencedirect.com/science/article/pii/0921452695000163>.
10. A. Ward and D. A. Broido, Intrinsic phonon relaxation times from first-principles studies of the thermal conductivities of si and ge, *Phys. Rev. B.* **81**, 085205 (Feb, 2010). doi: 10.1103/PhysRevB.81.085205. URL <https://link.aps.org/doi/10.1103/PhysRevB.81.085205>.
 11. A. Ward, D. A. Broido, D. A. Stewart, and G. Deinzer, Ab initio theory of the lattice thermal conductivity in diamond, *Phys. Rev. B.* **80**, 125203 (Sep, 2009). doi: 10.1103/PhysRevB.80.125203. URL <https://link.aps.org/doi/10.1103/PhysRevB.80.125203>.
 12. W. Li, N. Mingo, L. Lindsay, D. A. Broido, D. A. Stewart, and N. A. Katcho, Thermal conductivity of diamond nanowires from first principles, *Phys. Rev. B.* **85**, 195436 (May, 2012). doi: 10.1103/PhysRevB.85.195436. URL <https://link.aps.org/doi/10.1103/PhysRevB.85.195436>.
 13. D. A. Broido, L. Lindsay, and A. Ward, Thermal conductivity of diamond under extreme pressure: A first-principles study, *Phys. Rev. B.* **86**, 115203 (Sep, 2012). doi: 10.1103/PhysRevB.86.115203. URL <https://link.aps.org/doi/10.1103/PhysRevB.86.115203>.
 14. T. Luo, J. Garg, J. Shiomi, K. Esfarjani, and G. Chen, Gallium arsenide thermal conductivity and optical phonon relaxation times from first-principles calculations, *EPL (Europhysics Letters)*. **101**(1), 16001 (2013). URL <http://stacks.iop.org/0295-5075/101/i=1/a=16001>.
 15. L. Lindsay, D. A. Broido, and T. L. Reinecke, Ab initio thermal transport in compound semiconductors, *Phys. Rev. B.* **87**, 165201 (Apr, 2013). doi: 10.1103/PhysRevB.87.165201. URL <https://link.aps.org/doi/10.1103/PhysRevB.87.165201>.
 16. J. Shiomi, K. Esfarjani, and G. Chen, Thermal conductivity of half-heusler compounds from first-principles calculations, *Phys. Rev. B.* **84**, 104302 (Sep, 2011). doi: 10.1103/PhysRevB.84.104302. URL <https://link.aps.org/doi/10.1103/PhysRevB.84.104302>.
 17. X. Tang and J. Dong, Lattice thermal conductivity of mgo at conditions of earth's interior, *Proceedings of the National Academy of Sciences.* **107** (10), 4539-4543 (2010). ISSN 0027-8424. doi: 10.1073/pnas.0907194107. URL <http://www.pnas.org/content/107/10/4539>.
 18. H. Dekura and T. Tsuchiya, Ab initio lattice thermal conductivity of mgo from a complete solution of the linearized boltzmann transport equation, *Phys. Rev. B.* **95**, 184303 (May, 2017). doi: 10.1103/PhysRevB.95.184303. URL <https://link.aps.org/doi/10.1103/PhysRevB.95.184303>.
 19. H. Dekura, T. Tsuchiya, and J. Tsuchiya, Ab initio lattice thermal conductivity of mgsio₃ perovskite as found in earth's lower mantle, *Phys. Rev. Lett.* **110**, 025904 (Jan, 2013). doi: 10.1103/PhysRevLett.110.025904. URL <https://link.aps.org/doi/10.1103/PhysRevLett.110.025904>.
 20. Z. Tian, J. Garg, K. Esfarjani, T. Shiga, J. Shiomi, and G. Chen, Phonon conduction in pbse, pbte, and pbte_{1-x}se_x from first-principles calculations, *Phys. Rev. B.* **85**, 184303 (May, 2012). doi: 10.1103/PhysRevB.85.184303. URL <https://link.aps.org/doi/10.1103/PhysRevB.85.184303>.

21. W. Li, L. Lindsay, D. A. Broido, D. A. Stewart, and N. Mingo, Thermal conductivity of bulk and nanowire $\text{mg}_2\text{si}_x\text{sn}_{1-x}$ alloys from first principles, *Phys. Rev. B*, **86**, 174307 (Nov, 2012). doi: 10.1103/PhysRevB.86.174307. URL <https://link.aps.org/doi/10.1103/PhysRevB.86.174307>.
22. T. Pandey, C. A. Polanco, L. Lindsay, and D. S. Parker, Lattice thermal transport in $\text{La}_3\text{Cu}_3\text{X}_4$ compounds ($x = \text{P, As, Sb, Bi}$): Interplay of anharmonicity and scattering phase space, *Phys. Rev. B*, **95**, 224306 (Jun, 2017). doi: 10.1103/PhysRevB.95.224306. URL <https://link.aps.org/doi/10.1103/PhysRevB.95.224306>.
23. T. Pandey, D. S. Parker, and L. Lindsay, Ab initio phonon thermal transport in monolayer inSe, gas, and alloys, *Nanotechnology*, **28**(45), 455706 (2017). URL <http://stacks.iop.org/0957-4484/28/i=45/a=455706>.
24. L. Lindsay, D. A. Broido, and T. L. Reinecke, First-principles determination of ultrahigh thermal conductivity of boron arsenide: A competitor for diamond?, *Phys. Rev. Lett.* **111**, 025901 (Jul, 2013). doi: 10.1103/PhysRevLett.111.025901. URL <https://link.aps.org/doi/10.1103/PhysRevLett.111.025901>.
25. C. Shi and X. Luo, Characterization of lattice thermal transport in two-dimensional bas, bp, and bsb: A first-principles study, *arXiv:1811.05597v1* (2018).
26. G. P. Srivastava, *The Physics of Phonons*. CRC Press (1990).
27. T. Tadano, Y. Gohda, and S. Tsuneyuki, Impact of rattlers on thermal conductivity of a thermoelectric clathrate: A first-principles study, *Phys. Rev. Lett.* **114**, 095501 (Mar, 2015). doi: 10.1103/PhysRevLett.114.095501. URL <https://link.aps.org/doi/10.1103/PhysRevLett.114.095501>.
28. W. Li and N. Mingo, Ultralow lattice thermal conductivity of the fully filled skutterudite $\text{ybfe}_4\text{sb}_{12}$ due to the flat avoided-crossing filler modes, *Phys. Rev. B*, **91**, 144304 (Apr, 2015). doi: 10.1103/PhysRevB.91.144304. URL <https://link.aps.org/doi/10.1103/PhysRevB.91.144304>.
29. W. Li and N. Mingo, Thermal conductivity of fully filled skutterudites: Role of the filler, *Phys. Rev. B*, **89**, 184304 (May, 2014). doi: 10.1103/PhysRevB.89.184304. URL <https://link.aps.org/doi/10.1103/PhysRevB.89.184304>.
30. L. Lindsay and D. A. Broido, Three-phonon phase space and lattice thermal conductivity in semiconductors, *Journal of Physics: Condensed Matter*, **20** (16), 165209 (2008). URL <http://stacks.iop.org/0953-8984/20/i=16/a=165209>.
31. S.-i. Tamura, Isotope scattering of dispersive phonons in ge, *Phys. Rev. B*, **27**, 858–866 (Jan, 1983). doi: 10.1103/PhysRevB.27.858. URL <https://link.aps.org/doi/10.1103/PhysRevB.27.858>.
32. P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G. L. Chiarotti, M. Cococcioni, I. Dabo, A. D. Corso, S. de Gironcoli, S. Fabris, G. Fratesi, R. Gebauer, U. Gerstmann, C. Gougoussis, A. Kokalj, M. Lazzeri, L. Martin-Samos, N. Marzari, F. Mauri, R. Mazurek, S. Paolini, A. Pasquarello, L. Paulatto, C. Sbraccia, S. Scandolo, G. Sclauzero, A. P. Seitsonen, A. Smogunov, P. Umari, and R. M. Wentz-

- covitch, Quantum espresso: a modular and open-source software project for quantum simulations of materials, *Journal of Physics: Condensed Matter*. **21**(39), 395502 (2009). URL <http://stacks.iop.org/0953-8984/21/i=39/a=395502>.
33. W. Li, J. Carrete, N. A. Katcho, and N. Mingo, ShengBTE: a solver of the Boltzmann transport equation for phonons, *Comp. Phys. Commun.* **185**, 17471758 (2014). doi: 10.1016/j.cpc.2014.02.015.
 34. S. Sadasivam, M. K. Y. Chan, and P. Darancet, Theory of thermal relaxation of electrons in semiconductors, *Phys. Rev. Lett.* **119**, 136602 (Sep, 2017). doi: 10.1103/PhysRevLett.119.136602. URL <https://link.aps.org/doi/10.1103/PhysRevLett.119.136602>.
 35. Y. Wang, Z. Lu, and X. Ruan, First principles calculation of lattice thermal conductivity of metals considering phonon-phonon and phonon-electron scattering, *Journal of Applied Physics*. **119**(22), 225109 (2016). doi: 10.1063/1.4953366. URL <https://doi.org/10.1063/1.4953366>.
 36. A. Jain and A. J. H. McGaughey, Thermal transport by phonons and electrons in aluminum, silver, and gold from first principles, *Phys. Rev. B*. **93**, 081206 (Feb, 2016). doi: 10.1103/PhysRevB.93.081206. URL <https://link.aps.org/doi/10.1103/PhysRevB.93.081206>.
 37. L. Wei, P. K. Kuo, R. L. Thomas, T. R. Anthony, and W. F. Banholzer, Thermal conductivity of isotopically modified single crystal diamond, *Phys. Rev. Lett.* **70**, 3764–3767 (Jun, 1993). doi: 10.1103/PhysRevLett.70.3764. URL <https://link.aps.org/doi/10.1103/PhysRevLett.70.3764>.
 38. D. G. Onn, A. Witek, Y. Z. Qiu, T. R. Anthony, and W. F. Banholzer, Some aspects of the thermal conductivity of isotopically enriched diamond single crystals, *Phys. Rev. Lett.* **68**, 2806–2809 (May, 1992). doi: 10.1103/PhysRevLett.68.2806. URL <https://link.aps.org/doi/10.1103/PhysRevLett.68.2806>.
 39. J. R. Olson, R. O. Pohl, J. W. Vandersande, A. Zoltan, T. R. Anthony, and W. F. Banholzer, Thermal conductivity of diamond between 170 and 1200 K and the isotope effect, *Phys. Rev. B*. **47**, 14850–14856 (Jun, 1993). doi: 10.1103/PhysRevB.47.14850. URL <https://link.aps.org/doi/10.1103/PhysRevB.47.14850>.
 40. L. Lindsay, D. A. Broido, and T. L. Reinecke, Ab initio thermal transport in compound semiconductors, *Phys. Rev. B*. **87**, 165201 (Apr, 2013). doi: 10.1103/PhysRevB.87.165201. URL <https://link.aps.org/doi/10.1103/PhysRevB.87.165201>.
 41. E. G. Llesma, G. Mendes, C. Arguello, and R. Leite, Very high non-thermal equilibrium population of optical phonons in GaAs, *Solid State Communications*. **14**(11), 1195 – 1197 (1974). ISSN 0038-1098. doi: [https://doi.org/10.1016/0038-1098\(74\)90302-0](https://doi.org/10.1016/0038-1098(74)90302-0). URL <http://www.sciencedirect.com/science/article/pii/0038109874903020>.
 42. W. Pötz, Hot-phonon effects in bulk GaAs, *Phys. Rev. B*. **36**, 5016–5019 (Sep, 1987). doi: 10.1103/PhysRevB.36.5016. URL <https://link.aps.org/doi/10.1103/PhysRevB.36.5016>.
 43. M. N. Luckyanova, J. Garg, K. Esfarjani, A. Jandl, M. T. Bulsara, A. J.

- Schmidt, A. J. Minnich, S. Chen, M. S. Dresselhaus, Z. Ren, E. A. Fitzgerald, and G. Chen, Coherent phonon heat conduction in superlattices, *Science*. **338** (6109), 936–939 (2012). ISSN 0036-8075. doi: 10.1126/science.1225549. URL <http://science.sciencemag.org/content/338/6109/936>.
44. G. S. Nolas, J. L. Cohn, G. A. Slack, and S. B. Schujman, Semiconducting ge clathrates: Promising candidates for thermoelectric applications, *Applied Physics Letters*. **73**(2), 178–180 (1998). doi: 10.1063/1.121747. URL <https://doi.org/10.1063/1.121747>.
 45. G. S. Nolas, J. L. Cohn, and G. A. Slack, Effect of partial void filling on the lattice thermal conductivity of skutterudites, *Phys. Rev. B*. **58**, 164–170 (Jul, 1998). doi: 10.1103/PhysRevB.58.164. URL <https://link.aps.org/doi/10.1103/PhysRevB.58.164>.
 46. M. K. Jana, K. Pal, A. Warankar, P. Mandal, U. V. Waghmare, and K. Biswas, Intrinsic rattler-induced low thermal conductivity in zintl type tlnite2, *Journal of the American Chemical Society*. **139**(12), 4350–4353 (2017). doi: 10.1021/jacs.7b01434. URL <http://dx.doi.org/10.1021/jacs.7b01434>. PMID: 28263613.
 47. W. G. Zeier, A. Zevalkink, Z. M. Gibbs, G. Hautier, M. G. Kanatzidis, and G. J. Snyder, Thinking like a chemist: Intuition in thermoelectric materials, *Angewandte Chemie International Edition*. **55**(24), 6826–6841 (2016). ISSN 1521-3773. doi: 10.1002/anie.201508381. URL <http://dx.doi.org/10.1002/anie.201508381>.
 48. M. M. Koza, M. R. Johnson, R. Viennois, H. Mutka, L. Girard, and D. Ravot, Breakdown of phonon glass paradigm in la- and ce-filled fe4sb12 skutterudites, *Nature Materials*. **7**, 805 EP 810 (Aug, 2008). URL <https://doi.org/>.
 49. D. J. Voneshen, K. Refson, E. Borissenko, M. Krisch, A. Bosak, A. Piovano, E. Cemal, M. Enderle, M. J. Gutmann, M. Hoesch, M. Roger, L. Gannon, A. T. Boothroyd, S. Uthayakumar, D. G. Porter, and J. P. Goff, Suppression of thermal conductivity by rattling modes in thermoelectric sodium cobaltate, *Nature Materials*. **12**, 1028 EP – (Aug, 2013). URL <https://doi.org/10.1038/nmat3739>.
 50. E. S. Toberer, A. Zevalkink, and G. J. Snyder, Phonon engineering through crystal chemistry, *J. Mater. Chem.* **21**, 15843–15852 (2011). doi: 10.1039/C1JM11754H. URL <http://dx.doi.org/10.1039/C1JM11754H>.
 51. J. L. Cohn, G. S. Nolas, V. Fessatidis, T. H. Metcalf, and G. A. Slack, Glasslike heat conduction in high-mobility crystalline semiconductors, *Phys. Rev. Lett.* **82**, 779–782 (Jan, 1999). doi: 10.1103/PhysRevLett.82.779. URL <https://link.aps.org/doi/10.1103/PhysRevLett.82.779>.
 52. J. S. Tse, V. P. Shpakov, V. R. Belosludov, F. Trouw, Y. P. Handa, and W. Press, Coupling of localized guest vibrations with the lattice modes in clathrate hydrates, *EPL (Europhysics Letters)*. **54**(3), 354 (2001). URL <http://stacks.iop.org/0295-5075/54/i=3/a=354>.
 53. Y. Takasu, T. Hasegawa, N. Ogita, M. Udagawa, M. A. Avila, K. Suekuni, I. Ishii, T. Suzuki, and T. Takabatake, Dynamical properties of guest ions in the type-i clathrate compounds $X_{8ga_{16}ge_{30}}$ ($x = \text{Eu, Sr, Ba}$) investigated by raman scattering, *Phys. Rev. B*. **74**, 174303 (Nov, 2006). doi:

- 10.1103/PhysRevB.74.174303. URL <https://link.aps.org/doi/10.1103/PhysRevB.74.174303>.
54. M. A. Avila, K. Suekuni, K. Umeo, H. Fukuoka, S. Yamanaka, and T. Takabatake, Glasslike versus crystalline thermal conductivity in carrier-tuned $\text{BaGa}_{16}\text{X}_{30}$ clathrates ($x = \text{Ge}, \text{Sn}$), *Phys. Rev. B.* **74**, 125109 (Sep, 2006). doi: 10.1103/PhysRevB.74.125109. URL <https://link.aps.org/doi/10.1103/PhysRevB.74.125109>.
 55. T. Lay, J. Hernlund, and B. A. Buffett, Core-mantle boundary heat flow, *Nature Geoscience.* **1**, 25 EP – (Jan, 2008). URL <https://doi.org/10.1038/ngeo.2007.44>. Review Article.
 56. D. Gubbins, A. P. Willis, and B. Sreenivasan, Correlation of earths magnetic field with lower mantle thermal and seismic structure, *Physics of the Earth and Planetary Interiors.* **162**(3), 256 – 260 (2007). ISSN 0031-9201. doi: <https://doi.org/10.1016/j.pepi.2007.04.014>. URL <http://www.sciencedirect.com/science/article/pii/S0031920107000908>.
 57. G. M. Manthilake, N. de Koker, D. J. Frost, and C. A. McCammon, Lattice thermal conductivity of lower mantle minerals and heat flux from earth's core, *Proceedings of the National Academy of Sciences.* **108**(44), 17901–17904 (2011). ISSN 0027-8424. doi: 10.1073/pnas.1110594108. URL <http://www.pnas.org/content/108/44/17901>.
 58. A. F. Goncharov, S. S. Lobanov, X. Tan, G. T. Hohensee, D. G. Cahill, J.-F. Lin, S.-M. Thomas, T. Okuchi, and N. Tomioka, Experimental study of thermal conductivity at high pressures: Implications for the deep earths interior, *Physics of the Earth and Planetary Interiors.* **247**, 11 – 16 (2015). ISSN 0031-9201. doi: <https://doi.org/10.1016/j.pepi.2015.02.004>. URL <http://www.sciencedirect.com/science/article/pii/S0031920115000199>. Transport Properties of the Earths Core.
 59. K. Ohta, T. Yagi, K. Hirose, and Y. Ohishi, Thermal conductivity of ferropericlase in the earth's lower mantle, *Earth and Planetary Science Letters.* **465**, 29 – 37 (2017). ISSN 0012-821X. doi: <https://doi.org/10.1016/j.epsl.2017.02.030>. URL <http://www.sciencedirect.com/science/article/pii/S0012821X17300985>.
 60. A. M. Hofmeister, Mantle values of thermal conductivity and the geotherm from phonon lifetimes, *Science.* **283**(5408), 1699–1706 (1999). ISSN 0036-8075. doi: 10.1126/science.283.5408.1699. URL <http://science.sciencemag.org/content/283/5408/1699>.
 61. H. Aramberri, R. Ruruli, and J. Íñiguez, Thermal conductivity changes across a structural phase transition: The case of high-pressure silica, *Phys. Rev. B.* **96**, 195201 (Nov, 2017). doi: 10.1103/PhysRevB.96.195201. URL <https://link.aps.org/doi/10.1103/PhysRevB.96.195201>.
 62. A. F. Goncharov, P. Beck, V. V. Struzhkin, B. D. Haugen, and S. D. Jacobsen, Thermal conductivity of lower-mantle minerals, *Physics of the Earth and Planetary Interiors.* **174**(1), 24 – 32 (2009). ISSN 0031-9201. doi: <https://doi.org/10.1016/j.pepi.2008.07.033>. URL <http://www.sciencedirect.com/science/article/pii/S0031920108001945>. Advances in High Pressure Mineral Physics: from Deep Mantle to the Core.

63. D. A. Dalton, W.-P. Hsieh, G. T. Hohensee, D. G. Cahill, and A. F. Goncharov, Effect of mass disorder on the lattice thermal conductivity of mgo periclase under pressure, *Scientific Reports*. **3**, 2400 EP – (Aug, 2013). URL <https://doi.org/10.1038/srep02400>. Article.
64. J. M. Brown, Interpretation of the d zone at the base of the mantle: Dependence on assumed values of thermal conductivity, *Geophysical Research Letters*. **13**(13), 1509–1512 (1986). doi: 10.1029/GL013i013p01509. URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/GL013i013p01509>.
65. A. M. Hofmeister, Inference of high thermal transport in the lower mantle from laser-flash experiments and the damped harmonic oscillator model, *Physics of the Earth and Planetary Interiors*. **170**(3), 201 – 206 (2008). ISSN 0031-9201. doi: <https://doi.org/10.1016/j.pepi.2008.06.034>. URL <http://www.sciencedirect.com/science/article/pii/S0031920108002331>. Frontiers and Grand Challenges in Mineral Physics of the Deep Mantle.
66. R. Jeanloz and S. Morris, Temperature distribution in the crust and mantle, *Annual Review of Earth and Planetary Sciences*. **14**(1), 377–415 (1986). doi: 10.1146/annurev.ea.14.050186.002113. URL <https://doi.org/10.1146/annurev.ea.14.050186.002113>.
67. R. E. Cohen, Thermal conductivity of mgo at high pressures, *Rev High Pressure Sci Technol*. **7**, 160–162 (1998). doi: 10.4131/jshpreview.7.160.
68. N. de Koker, Thermal conductivity of mgo periclase from equilibrium first principles molecular dynamics, *Phys. Rev. Lett.* **103**, 125902 (Sep, 2009). doi: 10.1103/PhysRevLett.103.125902. URL <https://link.aps.org/doi/10.1103/PhysRevLett.103.125902>.
69. J. Badro, Spin transitions in mantle minerals, *Annual Review of Earth and Planetary Sciences*. **42**(1), 231–248 (2014). doi: 10.1146/annurev-earth-042711-105304. URL <https://doi.org/10.1146/annurev-earth-042711-105304>.
70. J.-F. Lin, S. Speziale, Z. Mao, and H. Marquardt, Effects of the electronic spin transitions of iron in lower mantle minerals: Implications for deep mantle geophysics and geochemistry, *Reviews of Geophysics*. **51**(2), 244–275 (2013). doi: 10.1002/rog.20010. URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/rog.20010>.
71. W.-P. Hsieh, F. Deschamps, T. Okuchi, and J.-F. Lin, Effects of iron on the lattice thermal conductivity of earth’s deep mantle and implications for mantle dynamics, *Proceedings of the National Academy of Sciences*. **115**(16), 4099–4104 (2018). ISSN 0027-8424. doi: 10.1073/pnas.1718557115. URL <http://www.pnas.org/content/115/16/4099>.
72. N. de Koker, Thermal conductivity of mgo periclase at high pressure: Implications for the d region, *Earth and Planetary Science Letters*. **292**(3), 392 – 398 (2010). ISSN 0012-821X. doi: <https://doi.org/10.1016/j.epsl.2010.02.011>. URL <http://www.sciencedirect.com/science/article/pii/S0012821X10001135>.
73. S. Stackhouse, L. Stixrude, and B. B. Karki, Thermal conductivity of periclase (mgo) from first principles, *Phys. Rev. Lett.* **104**, 208501 (May, 2010).

- doi: 10.1103/PhysRevLett.104.208501. URL <https://link.aps.org/doi/10.1103/PhysRevLett.104.208501>.
74. V. Haigis, M. Salanne, and S. Jahn, Thermal conductivity of mgo, mg-sio3 perovskite and post-perovskite in the earth's deep mantle, *Earth and Planetary Science Letters*. **355-356**, 102 – 108 (2012). ISSN 0012-821X. doi: <https://doi.org/10.1016/j.epsl.2012.09.002>. URL <http://www.sciencedirect.com/science/article/pii/S0012821X12004815>.
75. Y. Long and J. Chen, Theoretical study of the phononphonon scattering mechanism and the thermal conductive coefficients for energetic material, *Philosophical Magazine*. **97**(28), 2575–2595 (2017). doi: 10.1080/14786435.2017.1343962. URL <https://doi.org/10.1080/14786435.2017.1343962>.
76. S. Mukhopadhyay, D. S. Parker, B. C. Sales, A. A. Puretzky, M. A. McGuire, and L. Lindsay, Two-channel model for ultralow thermal conductivity of crystalline tl3vse4, *Science*. **360**(6396), 1455–1458 (2018). ISSN 0036-8075. doi: 10.1126/science.aar8072. URL <http://science.sciencemag.org/content/360/6396/1455>.
77. Y. Zeng and J. Dong, The fokker-planck equation for lattice vibration: Stochastic dynamics and thermal conductivity, *arXiv:1811.08492v1* (2018).